Michael Asemota

Arlington, TX 76010 | (469) 767 0976 | michael.asemotaa@gmail.com
GitHub Portfolio: https://ybmikey.github.io/Michael-A-Projects/index.html
Candidate for Engineering, seeking internships for 2026 & 2027

Education

University of Texas at Arlington
2023 - Present (Expected 2027)
- Industrial Engineering Major

Allen High School
2019 - 2023
- 3.8 Weighted GPA

Technical Skills
- CAD: SolidWorks, AutoCAD, Onshape, Fusion360, MATLAB
- Programming: C++, Python, Arduino
- Tools/Systems: Arduino Uno, Robotics, 3D Printing, Gimbal Systems
- Soft Skills: Six Sigma - White Belt, Lean Manufacturing, MS Office Applications (Excel, Word, etc.), DFM, DFA, FMEA, Fixture Design, Root Cause Analysis, Bottleneck Analysis

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Job Experience

Verlo (Technical Support)
2026 - Present
- Matched purchase requests to optimized mattress configurations using parameters such as foam density, and coil system architecture.
- Used spreadsheets to organize and analyze product data, improving collection workflows and efficiency.
- Leveraged customer and product data to identify process improvements in manufacturing and distribution efficiency.

Skechers (Associate Manager)
2022 - 2024
- Supervised daily operations for a high-volume retail store, leading 10–15 associates and increasing sales performance by 15–20% through structured coaching and performance tracking.
- Reduced stockouts by 25% by monitoring inventory and coordinating restocking using MS Office tools.
- Improved labor efficiency by 10–15% through sales data analysis and optimized scheduling, while implementing stockroom process improvements that reduced retrieval time by 30%.

Activities

NSBE (National Society of Black Engineers)
2023 - Present
- Participated in technical workshops and engineering events promoting diversity in STEM
- Engaged in collaborative engineering projects and networking opportunities with industry professionals
- Contributed to community outreach and professional development initiatives

Formula SAE
2023 - Present
- Contributed to the design, fabrication, and testing of high-performance vehicles in Formula SAE
- Applied knowledge in aerodynamics, mechanical systems, and electronics to improve vehicle performance
- Collaborated across sub-teams to troubleshoot design issues and optimize system integration

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Leadership

Robomasters (Gimbal Team Lead - 3D Modeling & Secretary)
2024 - Present
- Organized and led structured team meetings, managed technical documentation, and streamlined communication workflows to support cross-functional coordination.
- Served as liaison between internal teams and external stakeholders, including sponsors and competition organizers, ensuring clear requirements alignment and project tracking.
- Directed end-to-end gimbal system development, overseeing mechanical assembly, integration, and performance optimization.
- Guided system testing and calibration efforts, improving control accuracy and operational stability through iterative validation.
- Designed and assembled a precision gimbal mechanism incorporating high-torque motors and structural components, collaborating with teammates to refine control algorithms and ensure smooth, responsive motion.

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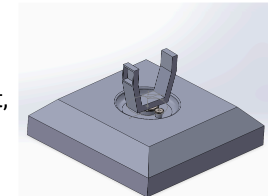
RoboMasters Standard Robot



What

- Contributed to the end-to-end mechanical design and system integration of a RoboMasters Standard Robot, including chassis structure, drivetrain architecture, and a dual-axis gimbal turret assembly.
- Collaborated cross-functionally with controls and electrical teams to integrate actuators, sensors, and targeting systems, ensuring mechanical compatibility and reliable motion performance.
- Developed and refined components using **SolidWorks** and **Fusion 360**, considering manufacturability, assembly sequence, and serviceability.

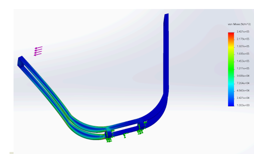
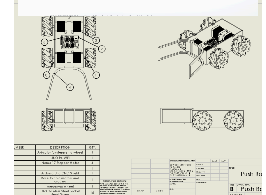
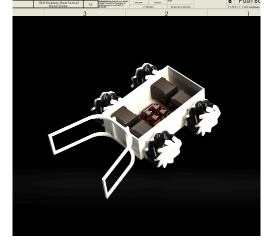
Why

- To develop a competition-ready robotic system capable of precise maneuvering, stable operation, and reliable targeting under dynamic loading conditions.
- To improve stability, accuracy, and repeatability of the turret system, enabling consistent targeting and shooting while the robot is in motion, while expanding the turret's effective range of motion.

Results

- Delivered a fully assembled, functional robot that competed in the RoboMasters 1v1 North America tournament, demonstrating stable navigation, responsive targeting, and a modular, scalable design suitable for testing, iteration, and future integration.

Arduino Mecanum Robot

What

- Engineered an omnidirectional robotic platform using Mecanum wheels, enabling controlled multi-axis motion and precise positioning.
- Programmed motion control algorithms in **C++ (Arduino Uno R4)** to achieve smooth acceleration, accurate direction control, and repeatable object manipulation.
- Designed the mechanical system using **SolidWorks** and **Onshape**, accounting for load distribution, wheel alignment, and drivetrain efficiency.

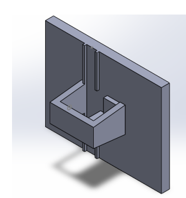
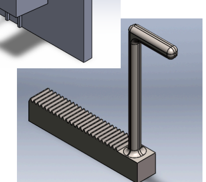
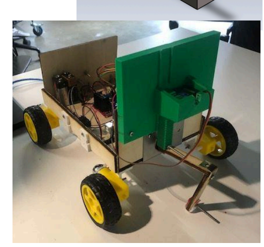
Why

- To design a system capable of efficient, precise motion within a constrained workspace, emphasizing controllability, repeatability, and system stability.
- To apply problem-solving and optimization techniques to overcome mechanical and control challenges inherent in omnidirectional motion systems.

Results

- Successfully completed the task in under 30 seconds, demonstrating high positional accuracy, reliable control, and consistent system performance.

Arduino Based Robot

What

- Designed and built a custom robotic system utilizing an Arduino microcontroller and a rack-and-pinion mechanism to perform a precision object transfer operation.
- Implemented **C++** control logic to manage actuator timing and sequencing, ensuring consistent and repeatable mechanical motion.
- Modeled the system in **SolidWorks**, focusing on mechanical accuracy, alignment, and tolerance control.

Why

- To solve a high-precision mechanical challenge requiring accurate motion control and strict adherence to spatial constraints without violating a defined boundary.
- To emphasize repeatability, mechanical reliability, and process consistency over trial-and-error solutions.

Results

- Achieved the objective flawlessly in under 30 seconds, with no faults or errors, demonstrating robust mechanical design and dependable control logic.